

EX NO:4

Design of Low Pass Filter using Windowing Techniques

4.1 : Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique.

Program (Matlab Code):

```
clc; clear; close all;

%% --- Input Parameters ---fp = 1000;

% Passband cutoff (Hz)

fs = 4000;      % Sampling frequency (Hz)

N = 50; % Filter order (even number preferred) wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- 1. FIR Low Pass using Rectangular Window ---b_rect = fir1(N, wc, 'low', rectwin(N+1));
disp('--- Rectangular Window Coefficients ---'); disp(b_rect);

%% --- 2. FIR Low Pass using Hamming Window ---b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---'); disp(b_hamm);

%% --- 3. FIR Low Pass using Hanning Window ---b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---'); disp(b_hann);

%% --- Plot Frequency Responses ---figure;

freqz(b_rect,1);

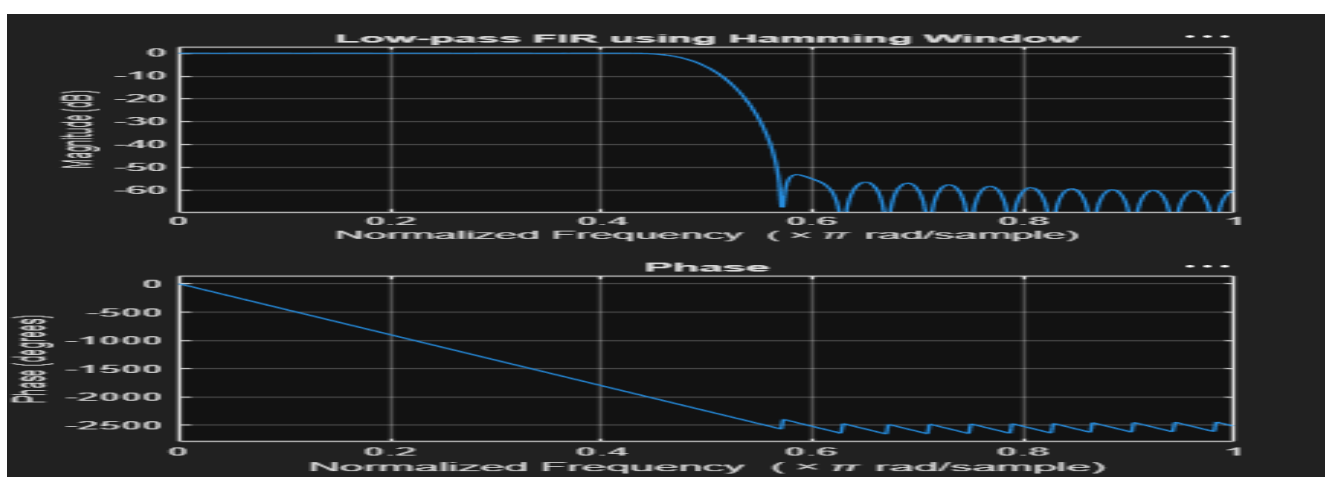
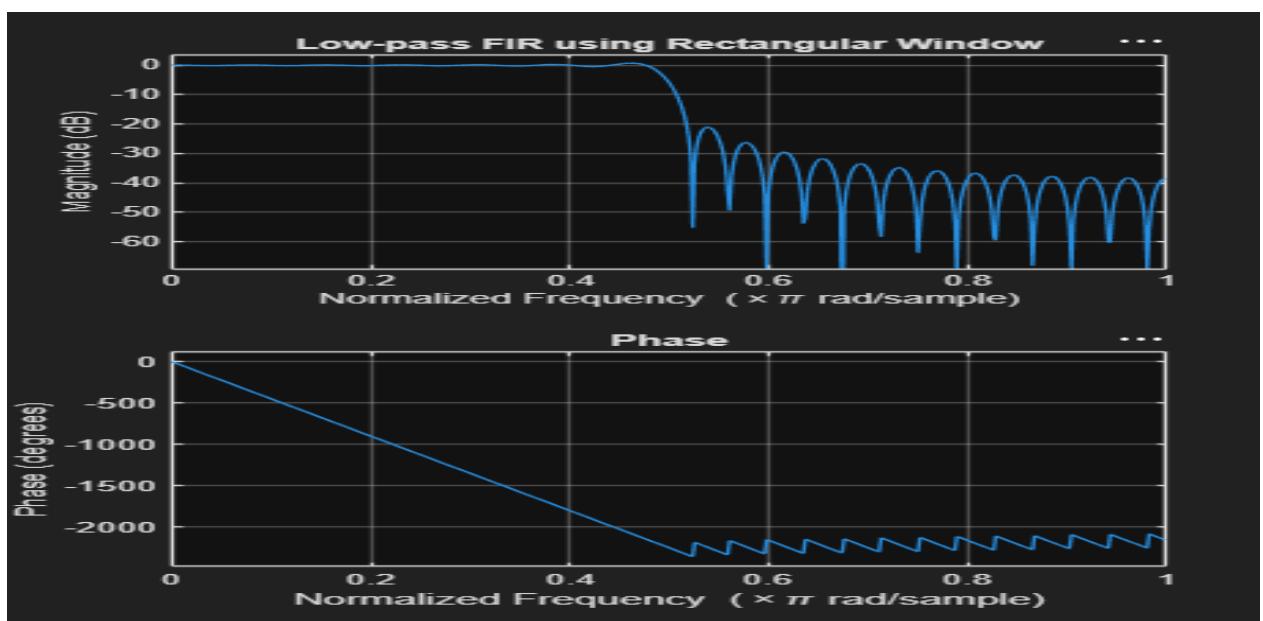
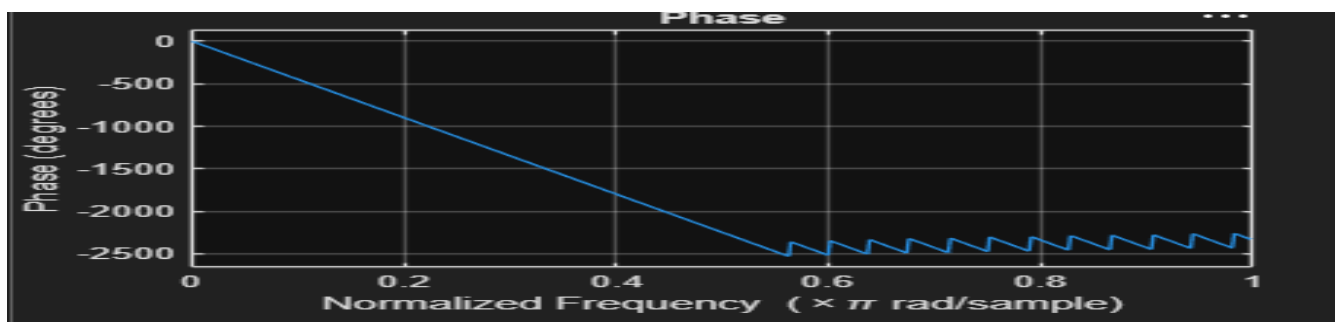
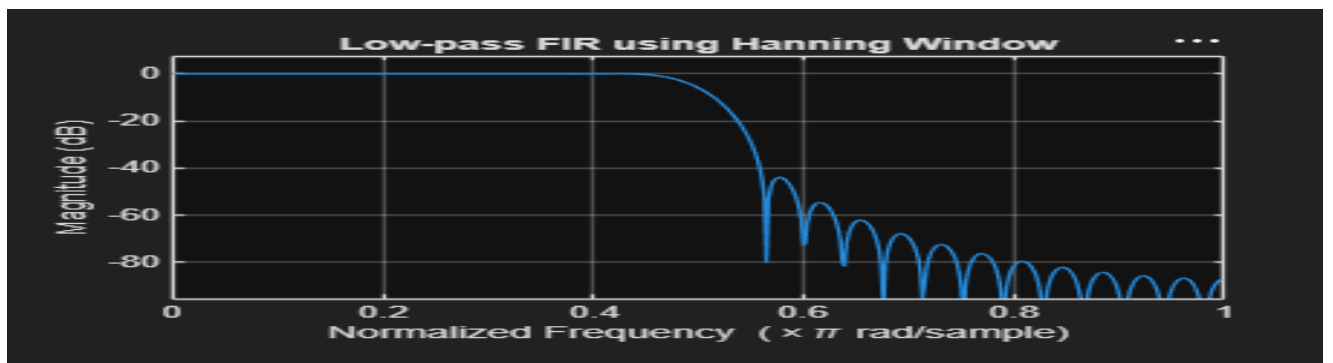
title('Low-pass FIR using Rectangular Window'); figure;

freqz(b_hamm,1);

title('Low-pass FIR using Hamming Window'); figure;

freqz(b_hann,1);

title('Low-pass FIR using Hanning Window');
```



Output

Rectangular Window Coefficients ---

0.01260
-0.01370
0.01500
-0.01660
0.01850
-0.02100
0.02420
-0.02860
0.03490
-0.04490
0.06290
-0.10480
0.3145
0.4940
0.31450
-0.10480
0.06290
-0.04490
0.03490
-0.02860
0.02420
-0.02100
0.01850
-0.01660
0.01500
-0.01370
0.0126

--- Hamming Window Coefficients ---

0.0010
-0.0013
0.0021
-0.0034
0.0055
-0.0084
0.0125
-0.0181
0.0260
-0.0379
0.0580
-0.1026
0.3168

0.4995

0.3168

-0.1026

Hanning Window Coefficients ---

0.0000

0

-0.0004

0

0.0013

0

-0.0028

0

0.0050

0

-0.0081

0

0.0122

0

-0.0179

0

0.0259

0

-0.0378

0

0.0580

0

-0.1027

0

0.3172

0.5000

0.3172

0

-0.1027

0

0.0580

0

-0.0378

0

0.0259

0

-0.0179

0

0.0122

0

-0.0081

0

0.0050

0

-0.0028

0

0.0013

0

-0.0004

0

0.0000

4.2: Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique

Program (Matlab Code):

```
clc;
clear;
close all;

%% --- Input Parameters ---
fp = 1000;    % Passband cutoff (Hz)
fs = 4000;    % Sampling frequency (Hz)
N = 50;       % Filter order (even number preferred)
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
disp(b_hamm);

%% --- Plot Frequency Responses ---
figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');
```

Output:

```
Hamming Window Coefficients ---
    0.0010
         0
   -0.0013
         0
    0.0021
         0
   -0.0034
         0
    0.0055
         0
   -0.0084
         0
    0.0125
```

0
-0.0181
0
0.0260
0
-0.0379
0
0.0580
0
-0.1026
0
0.3168
0.4995
0.3168
0
-0.1026
0
0.0580
0
-0.0379
0
0.0260
0
-0.0181
0
0.0125
0
-0.0084
0
0.0055
0
-0.0034
0
0.0021
0
-0.0013
0
0.0010

